

RAGHAVAN VAIDHYARAMAN

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Education

Columbia University, New York, NY | Fall 2026

Master of Science in Computer Science (Pathway: Machine Learning) | Expected Dec 2027

Coursework: Artificial Intelligence, Multimodal Data Management with AI Agents, Distributed Systems, Algorithms

Vellore Institute of Technology, Vellore | Jul 2020 - Aug 2024

B.Tech in Computer Science Engineering, Specialization in Data Science | CGPA: 8.54/10.00

Coursework: Mathematical Modeling and Programming for Data Science, NLP, Cloud & Database Systems, Statistics

Experience

Databricks | Associate Spark Technical Solutions Engineer | Bangalore, India | Jul 2024 - Jul 2026

- Optimized Spark and Delta pipelines across 50+ enterprise workloads; improved runtimes up to 80% and reduced cloud spend by ~40% through code optimization, partitioning, join tuning, and caching.
- Recognized as an SME in Apache Spark, Delta Lake, and MLflow; mentored 8 engineers and authored 20+ in depth technical articles for global enablement. Developed agent skills for AI-assisted workflows in Claude Code.
- Secured 1st runner-up, Databricks Global Hackathon 2025: built Genie AI, a voice-based assistant prototype.
- Delivered 2 nationwide Spark tech-talks and 10+ community solutions.

Databricks | Intern Trainee Engineer | Bangalore, India | Jan 2024 - Jun 2024

- Completed advanced labs in Spark, Delta, Structured Streaming, and Auto Loader; authored 15 reusable notebooks adopted for internal enablement.

Ideas2IT | Data Science Intern | Chennai, India | May 2023 - Jul 2023

- Built a GPT-powered email campaign generator to match prospects with relevant case studies, optimizing token usage using embedding-based ANN retrieval and fine-tuning. Increased reply rates by 40% for a 2,500-user base.

Selected Projects

- Algorithmic Trading Signal Generator** 🤖 (Data Engineering, Spark, MLflow): Built an end-to-end Databricks ML pipeline from ingestion to model serving, generating macroeconomic signals for S&P 500 stocks at 87.5% accuracy.
- LangChain Food Recommender** 🍽️ (LangChain, FAISS, OpenAI, Python): Built a RAG assistant over recipe PDFs with vector search; achieved 93% retrieval accuracy using LLM-as-judge evaluation.
- Multilingual Voice-to-SQL Assistant** (Whisper, Databricks Genie, Delta Lake): Built a BI prototype supporting 50+ languages; transcribing speech to English, generated SQL from it, and executed governed queries over Delta Lake.

Research & Publications

Vaidhyaraman, R., Modi, A., & Kaliappan, J. (2026). *Nonlinear classification on MiniRocket architecture for fish freshness prognostics*. Springer LNNS, Vol. 1749, pp. 277–288. DOI: 10.1007/978-3-032-13003-7_27

- Extended MiniRocket for multivariate nonlinear sensor time series and optimized inference for near real-time freshness monitoring, strengthening supply chain quality analysis.

Vaidhyaraman, R., S. J. S., & S. A. M. S. A. (2024). *BERT based Citation Recommender and Impactful Nodes Identifier in RDF Knowledge Graphs*. ICSCSA (IEEE), Salem, India, pp. 441–448. DOI: 10.1109/ICSCSA64454.2024.00077

- Built an RDF knowledge graph and BERT based citation ranking; combined embeddings with graph centrality techniques to surface influential papers, improve citation discovery, and accelerate literature review workflows.

Skills

- Programming:** Python, SQL, Scala, Java, C++, JavaScript, HTML/CSS, SPARQL
- AI & Machine Learning:** PyTorch, scikit-learn, pandas, Hugging Face Transformers, LangChain, RAG, Embeddings, FAISS, Fine-Tuning, LLM Evaluation, Model Serving, Vector Database, MLflow
- Data Systems:** Apache Spark, PySpark, Delta Lake, Structured Streaming, Auto Loader, PostgreSQL
- Cloud & Engineering:** Databricks, AWS, Azure, Docker, Kubernetes, Linux, Git, REST APIs, Vercel
- Certifications** 📜: Databricks Certified Associate Developer for Apache Spark; Databricks Certified Machine Learning Engineer Associate; Databricks Certified Generative AI Engineer Associate; IBM Certified AI Analyst
- Leadership:** PR Head, Youth Red Cross (coordinated quarterly blood donation drives with 4 hospitals); President, Becoming I Foundation (taught STEM courses to 100+ students across 8 schools)